

Mehrdad Momeni Zadeh

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EDUCATION

Simon Fraser University

Bachelor of Science in Computer Science

Burnaby, BC

Expected Sep 2027

- **GPA:** 3.4/4.33
- **Awards:** Dean's Honour Roll (Summer 2024, Summer 2025)
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Design, Database Systems, Computer Systems, Operating Systems, Software Engineering, Distributed Systems, Machine Learning, Deep Learning, Computer Networks

TECHNICAL SKILLS

Languages: Java, Python, Go, C++, TypeScript, SQL, Kotlin

Backend & Web: Spring Boot, Django, Express.js, React, Hibernate, REST APIs, JWT

Data & ML: PyTorch, scikit-learn, NumPy, Pandas, PostgreSQL, MySQL, MongoDB

DevOps & Tools: Docker, Git, AWS (EC2, S3), GCP, CI/CD, Linux, JUnit, pytest, Mockito

WORK EXPERIENCE

Research Assistant – DeLTA Lab

Simon Fraser University ([Project](#) — [Code](#))

Burnaby, BC

Jan 2025 – Aug 2025

- Developed and deployed ML training pipelines using Docker and GCP, reducing setup time by **40%**.
- Implemented CI/CD workflows for a **6-person research team**, streamlining code integration processes.
- Optimized data preprocessing scripts in Python, improving model training speed by **2.5×** on large datasets.
- Collaborated with graduate students to design automated experiments using PyTorch and WandB.
- Co-authored research paper on neural network optimization accepted to **ICLR 2026** conference.

TECHNICAL PROJECTS

Event Hub — Full-stack event ticketing platform for university organizations

[GitHub](#)

Tech: Java, Spring Boot, PostgreSQL, React, JWT, REST APIs

- Deployed for campus events, managing **200+ student registrations** with automated confirmation emails.
- Implemented secure authentication using Spring Security with **JWT tokens** and role-based access control.
- Collaborated in a **team of 3** to design REST APIs and optimize database queries for faster load times.

Micro Gopher — Distributed task processing system using Go microservices

[GitHub](#)

Tech: Go, Docker, PostgreSQL, MongoDB, gRPC, RabbitMQ

- Refactored monolithic application into **4 Go microservices** improving fault isolation and scalability.
- Developed auth, logging, and notification services communicating via **gRPC** with message queuing.
- Containerized services with Docker, achieving **500+ requests/minute** throughput in load testing.

Galactic Blitz — 2D space shooter with custom C++ game engine

[GitHub](#)

Tech: C++14, SFML, Box2D, CMake

- Built custom game engine from scratch supporting **real-time physics** and particle effects.
- Achieved stable **60 FPS** gameplay through optimized rendering pipeline and spatial partitioning.
- Implemented **entity-component architecture** allowing easy addition of new game mechanics.

Neuro Driver — AI-powered driving simulator with evolutionary learning

[GitHub](#)

Tech: Python, PyGame, NumPy, Genetic Algorithms

- Trained neural network agents reducing collision rate by **35%** over **50 generations** of evolution.
- Implemented fitness-based selection achieving **45% improvement** in track completion rates.
- Created visualization dashboard displaying real-time neural network decisions and performance metrics.